

§ 1. The Product Topology and the Tychonoff Theorem

§ 2. The Urysohn Lemma

§ 3. The Baire Category Theorem

Problem 3.1. Let X equal the countable union $\bigcup B_n$. Show that if X is a nonempty Baire space, at least one of the sets $\overline{B_n}$ has a nonempty interior.

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. ■

Problem 3.2. The Baire Category Theorem implies that $\mathbb{R}$ cannot be written as the countable union of closed subsets having empty interiors. Show this fails if the sets are not required to be closed.

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. ■

Problem 3.3. Show that the irrationals are a Baire space.

Proof. We will first show that any subset of a complete metric space which is a G_δ set is a Baire space. ■

Problem 3.4. Let (M, d) be a complete metric space with no isolated points. Prove that M is uncountable.

Proof. Suppose towards contradiction M is countable. Then we can write $M = \bigcup_{i=1}^{\infty} \{x_i\}$, where $\{x_i\}_{i=1}^{\infty}$ is an enumeration of M . Observe that:

$$\begin{aligned} \bigcap_{i=1}^{\infty} \{x_i\}^c &= \left(\bigcup_{i=1}^{\infty} \{x_i\} \right)^c \\ &= M^c \\ &= \emptyset. \end{aligned}$$

Since M has no isolated points, each $\{x_i\}$ is closed and nowhere dense. Hence each $\{x_i\}^c$ is open and dense. But surely from the above string of equalities $\bigcap_{i=1}^{\infty} \{x_i\}^c$ is not dense, hence M is not a Baire Space. This contradicts the Baire Category Theorem, as every complete metric space is also a Baire space. It must be the case that M is uncountable. ■

Problem 3.5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function, and suppose that for each $x \in \mathbb{R}$, there exists a natural number n such that $f^{(n)}(x) = 0$. For each fixed $n \in \mathbb{N}$, consider the set

$$F_n := \{x \in \mathbb{R} \mid f^{(n)}(x) = 0\}$$

and let $I := [a, b]$.

(a) Show that $I = \bigcup_{n=0}^{\infty} (I \cap F_n)$.

(b) Show that there is a nonempty open interval $A \subseteq I$ where f is a polynomial.

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. ■